

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

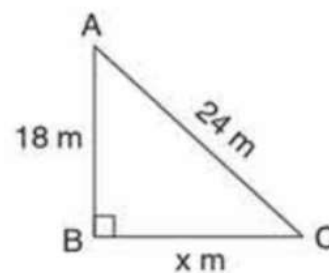
Mathematics

1. A cylindrical water tank has an internal radius of **3.5 m** and height **4 m**. The outer curved surface is to be painted at a rate of **₹18 per m²**. Ignoring thickness, what will be the **total cost of painting**?



- (A) ₹1386 (B) ₹1584
(C) ₹1764 (D) ₹1848
2. The average marks of **30 students** is 62 and that of **20 students** is 68. Find the **combined mean**.
- (A) 64.4 (B) 65.0
(C) 66.4 (D) 67.2
3. A ladder is placed against a vertical wall. The foot of the ladder is at point C on the

ground, the wall meets the ground at B, and the top of the ladder touches the wall at A. The height of the wall (A(B) is 18 m and the length of the ladder (A(C) is 24 m, as shown in the figure. **Find the distance of the foot of the ladder from the wall?**



- (A) 12 m (B) $\sqrt{252}$ m
(C) 18 m (D) $\sqrt{200}$
4. A solid iron sphere of radius 14 cm is melted and recast into 8 identical smaller spheres. **What is the radius of each smaller sphere?**
- (A) 5 cm (B) 6 cm
(C) 7 cm (D) 8 cm

5. **Statement-I:** The equation $x^2 + 6x + 5 = 0$ has no real roots.

Statement-II: If the discriminant of a quadratic equation is negative, its roots are non-real.

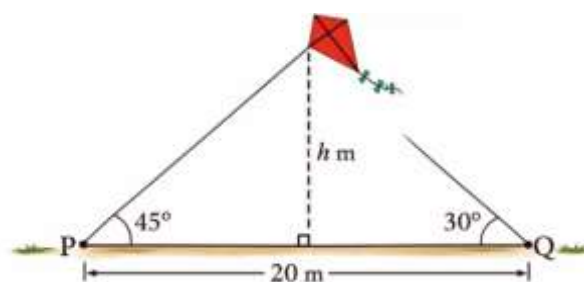
- (A) Both Statement-I and Statement-II are true and Statement-II is the correct explanation of Statement-I
(B) Both Statement-I and Statement-II are true but Statement-II is NOT the correct explanation of Statement-I
(C) Statement-I is true but Statement-II is false
(D) Statement-I is false but Statement-II is true
6. Two bells ring at regular intervals. The first bell rings every 48 seconds and the second every 72 seconds. If both bells ring together at 10:00:00 a.m., at **what time will they ring together again for the first time?**
- (A) 10:02:24 a.m. (B) 10:01:12 a.m.
(C) 10:03:36 a.m. (D) 10:00:36 a.m.
7. A car manufacturer plans to increase production in a fixed pattern. In the first month, they produce 2,000 cars. Every month, the production increases by the same fixed number. If the total production in the first 12 months is 37,200 cars, what is the monthly increase in production?

- (A) 150 (B) 200
(C) 250 (D) 300

8. A flagstaff stands on the top of a tower. From a point on the ground, the angles of elevation of the **top and bottom of the flagstaff** are **60°** and **45°** respectively. If the height of the tower is **20 m**, the height of the flagstaff is:

- (A) 5 m (B) $10(\sqrt{3} + 1)$
(C) 15 m (D) $20(\sqrt{3} - 1)$ m

9. A kite is flying at a constant height. The angles of elevation of the kite from two points **P** and **Q** on the same horizontal line and on the same side of the kite are **45°** and **30°** respectively. If $PQ = 20$ m, then the height of the kite is:



- (A) 10 m (B) $10(\sqrt{3} + 1)$ m
(C) 20 m (D) $20\sqrt{3}$ m

10. If the mean of a distribution is 50 and the mode is 44, **find the median.**

- (A) 46 (B) 47
(C) 48 (D) 49

11. A rectangular playground has dimensions 105 m and 140 m. It is to be divided into square plots of equal size without any remainder.

What is the **maximum possible area** of each square plot?

- (A) 35 m^2 (B) 49 m^2
(C) 1225 m^2 (D) 196 m^2

12. The seating arrangement in an auditorium has 18 seats in the first row. Each successive row has 4 more seats than the previous one. How many seats are there in the first 25 rows together?

- (A) 1,975 (B) 1,650
(C) 2,025 (D) 2,050

13. A ladder makes an angle θ with the ground such that $\sin \theta = \frac{5}{13}$. If the ladder is **13 m long**, the height reached by the ladder is:

- (A) 5 m (B) 12 m
(C) 8 m (D) 10 m

14. The sum of two numbers is 45. If one-fourth of the first number is added to one-third of the second number, the result is 14. Find the first number.

- (A) 15 (B) 12
(C) 21 (D) 24

15. A solid is made of a cylinder surmounted by a cone of same base radius **3.5 cm**. Cylinder height = 10 cm, cone height = 6 cm. Find total volume.

- (A) 577.5 cm^3 (B) 462 cm^3
(C) 770 cm^3 (D) 808.5 cm^3

16. The following data represents the daily travel distance (in km) of 100 delivery agents:

Distance (km)	Frequency
0–10	12
10–20	18
20–30	40
30–40	20
40–50	10

What is the median distance travelled?

- (A) 22.5 km (B) 25 km
(C) 27.5 km (D) 30 km

17. A farmer wants to pack 306 apples and 657 oranges in boxes such that each box has equal number of apples and oranges, and no fruit is left out. What is the maximum number of boxes possible?

- (A) 9 (B) 18
(C) 27 (D) 36

18. A machine depreciates in value every year following an arithmetic pattern. If its value after 4 years is ₹52,000 and after 10 years is ₹34,000, find its original value.

- (A) ₹64,000 (B) ₹62,000
(C) ₹63,500 (D) ₹65,000

19. From the top of a lighthouse **50 m high**, the angles of depression of two ships in the same straight line are **30°** and **60°** . **The distance between the two ships is:**

- (A) 50 m (B) $50\sqrt{3}$ m
(C) $\frac{100}{\sqrt{3}}$ m (D) $25\sqrt{3}$ m

20. A password is formed using **one digit** selected at random from the digits 1 to 9.

What is the probability that the selected digit is **a prime number or a multiple of 3**?

- (A) $\frac{5}{9}$ (B) $\frac{6}{8}$
(C) $\frac{2}{3}$ (D) $\frac{7}{9}$

21. From the top of a building **30 m high**, the angles of depression of the top and foot of a tower are **30°** and **60°** respectively.

The height of the tower is:

- (A) 20 m (B) 30 m
(C) 40 m (D) 50 m

22. The difference between the cost of 3 notebooks and 2 pens is ₹16. The total cost of 5 notebooks and 4 pens is ₹94.

What is the cost of one notebook?

- (A) ₹10 (B) ₹11.45
(C) ₹14.5 (D) ₹16

23. The product of two positive integers is 2700. If their HCF is 30 and one number

has exactly 3 prime factors (counting repetition), **then the other number is:**

- (A) 90 (B) 150
(C) 180 (D) 300

24. A solid cylinder of radius **4 cm** and height **7 cm** is melted and drawn into a wire of radius **0.5 cm**. Find the **length of the wire**.

- (A) 448 cm (B) 560 cm
(C) 896 cm (D) 1120 cm

25. A suspension bridge cable is modeled by the quadratic polynomial $p(x) = x^2 - 9x + 14$, where x represents the horizontal distance (in metres) from one end. The points where the cable touches the road are the zeroes of the polynomial. Which of the following correctly represents the distances at **which the cable touches the road?**

- (A) 2 m and 7 m
(B) -2 m and -7 m
(C) 9 m and 14 m
(D) 1 m and 14 m

26. A sector of a circle has radius **14 cm** and angle **90°** . Find the **area of the corresponding segment**

(Use $\pi = \frac{22}{7}$).

- (A) 154 cm^2 (B) 308 cm^2
(C) 158 cm^2 (D) 56 cm^2

27. Five years ago, the age of a father was three times the age of his son. After five years, the sum of their ages will be 70 years.

What is the present age of the father?

- (A) 40 years (B) 42.5 years
(C) 45 years (D) 50.5 years

28. Three numbers are in arithmetic progression. If the sum of the first two numbers is 10 and the sum of the last two numbers is 14, **what is the middle number?**

- (A) 5 (B) 6
(C) 7 (D) 8

29. A company tests a machine using a polynomial performance model $p(x) = 2x^3 - 5x^2 + 4x - 7$. If the machine is tested at setting $x = 1$, the remainder obtained on division by $x - 1$ indicates efficiency loss.

What is the efficiency loss?

- (A) -6 (B) -4
(C) -3 (D) -1

30. In a triangle, the difference between the largest and smallest angle is 40° , and the third angle is 60° .

Find the largest angle.

- (A) 80° (B) 70°
(C) 90° (D) 100°

31. The following table shows the number of hours students spent on online learning in a week.

Hours (x)	Number of Students (f)
2	5
4	9
6	12
8	10
10	4

Find the **mean number of hours** spent by a student.

- (A) 5.95 (B) 6.0
(C) 6.2 (D) 6.8

32. A circular park has a lamp post fixed at point **P** outside the park. From **P**, a straight road touches the park at **T**. The distance of **P** from the center **O** of the park is **26 m** and the radius of the park is **10 m**. What is the **length of the road PT**?

- (A) 20 m (B) 22 m
(C) 24 m (D) 16 m

33. If the zeroes of a quadratic polynomial differ by 4 and their product is 5, **then the polynomial is:**

- (A) $x^2 - 4x + 5$ (B) $x^2 - 6x + 5$
(C) $x^2 + 4x - 5$ (D) $x^2 - 5x + 6$

34. The sum of first 15 terms of an A.P. equals the sum of first 20 terms of another A.P. having the same common difference. If their first terms are 6 and 1 respectively, find the common difference.

(A) 2 (B) $\frac{4}{19}$
(C) $\frac{14}{17}$ (D) 5

35. A rectangular box has dimensions **40 cm × 25 cm × 10 cm**. How much **cardboard area** is needed to make **5 such closed boxes**?

(A) 11,500 cm² (B) 13,000 cm²
(C) 15,000 cm² (D) 16,500 cm²

36. Match Column I (Situation) with Column II (Correct Probability) and choose the correct option.

Column I	Column II
(a) Probability of getting a 'Tail' when a fair coin is tossed	(p) $1/6$
(b) Probability of getting a composite number (4, 6) on a single die throw	(q) $1/2$
(c) Probability of getting the number '5' on a single die throw	(r) $1/4$
(d) Probability of getting two Heads when two coins are tossed simultaneously	(s) $1/3$

(A) a–p, b–q, c–r, d–s
(B) a–q, b–s, c–p, d–r
(C) a–p, b–s, c–r, d–q
(D) a–q, b–p, c–s, d–r

37. If $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$ and $\frac{2}{x} - \frac{1}{y} = \frac{1}{3}$ then **find x**.

(A) 6 (B) 12
(C) 18 (D) 24

38. A satellite signal tower is installed at a point dividing the line joining A(4, –2) and B(10, 8) internally in the ratio 2:3.

Find the coordinates of the tower.

(A) (6.4, 2) (B) (7, 4)
(C) (8.5, 5) (D) (5, 1)

39. A flagstaff fixed vertically on level ground casts a shadow of 3 m at a certain time of the day. At the same moment, a nearby tower casts a shadow of 15 m. If the height of the flagstaff is 2.4 m, assuming the sun's rays make the same angle with the ground, **what is the height of the tower?**

(A) 10 m (B) 12 m
(C) 14 m (D) 15 m

40. A train travels a distance of **400 km** at a certain speed. If the speed were **10 km/h more**, it would take **2 hours less** for the same journey. **Find the original speed.**

(A) 30 km/h (B) 40 km/h
(C) 50 km/h (D) 55 km/h

41. A triangle has angles **2x°**, **(3x + 10)°**, and **(x + 20)°**. **What type of triangle is it?**

(A) Acute (B) Right-angled
(C) Obtuse (D) Equilateral

42. A security camera is placed at point $P(x, y)$ such that it is equidistant from gates $A(3, 7)$ and $B(7, 3)$.

Which equation represents the possible locations of P ?

- (A) $x + y = 10$ (B) $x - y = 0$
(C) $x^2 + y^2 = 58$ (D) $x - y = 4$

43. The product of two consecutive positive integers is **600**.

What is the larger integer?

- (A) 24 (B) 25
(C) 26 (D) 30

44. The areas of two similar triangles are 64 cm^2 and 100 cm^2 . If the smaller triangle has a side of 8 cm,

find the corresponding side of the larger triangle.

- (A) 10 cm (B) 12 cm
(C) 14 cm (D) 16 cm

45. A school conducts a lucky draw where one token is picked at random from a box containing **12 red tokens, 8 blue tokens, and 10 green tokens**.

A student wins a prize if the token drawn is **either blue or green**.

What is the probability that the student wins the prize?

- (A) $3/5$ (B) $9/15$
(C) $3/10$ (D) $18/35$

46. Two mobile towers are located at $(-6, 4)$ and $(2, -2)$. A relay station must be placed exactly midway. Where should it be installed?

- (A) $(-2, 1)$ (B) $(-4, 2)$
(C) $(-1, 1)$ (D) $(-3, 0)$

47. A triangular park has sides in the ratio **3 : 4 : 5**. If the perimeter of the park is **240 m**, a walking track is built along the **longest side only**. What is the length of the walking track?

- (A) 90 m (B) 100 m
(C) 120 m (D) 80 m

48. In a factory, 40 workers earn an average daily wage of ₹520 and another 60 workers earn an average daily wage of ₹680. If all workers are considered together, **what is the overall average daily wage?**

- (A) ₹616 (B) ₹620
(C) ₹640 (D) ₹66

49. A ball is thrown upward from the top of a building. Its height h (in metres) after t seconds is given by $h = -5t^2 + 20t + 25$. After how many seconds does the ball hit the ground?

- (A) 1 s (B) 3 s
(C) 5 s (D) 7 s

50. A box contains 4 red, 5 blue, and 3 green balls. Two balls are drawn without replacement. What is the probability that **both balls are of the same colour?**

- (A) $19/66$ (B) $11/66$
(C) $5/66$ (D) $1/11$

